

$$1. \begin{cases} \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, & x, y \text{ 在 } \Omega \text{ 之内} \\ \frac{\partial u}{\partial n} \Big|_S = 0 \end{cases}$$

$$2. \begin{cases} \frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2}, & 0 < x < 1, t > 0 \\ u|_{x=0} = 0, & u|_{x=1} = 0 \\ u|_{t=0} = f(x), & \frac{\partial u}{\partial t} \Big|_{t=0} = g(x) \end{cases}$$

$$3. \begin{cases} \frac{\partial u}{\partial t} = a^2 \left( \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} \right), & 0 < r < R, t > 0 \\ u|_{r=R} = 0 \\ u|_{t=0} = 1 - \left( \frac{r}{R} \right)^2 \end{cases}$$

二. 解: 根据边界条件知特殊函数为  $\cos \frac{n\pi}{L}x$

$$\therefore \text{假设 } u(x, t) = \sum_{n=0}^{\infty} u_n(t) \cos \frac{n\pi}{L}x \quad \text{--- ①}$$

将①式代入原方程得

$$\sum_{n=0}^{\infty} [u_n''(t) + \frac{n^2\pi^2}{L^2} u_n(t)] \cos \frac{n\pi}{L}x = \cos \frac{2\pi}{L}x$$

$$\text{观察得 } \uparrow \quad u_2''(t) + \frac{4\pi^2}{L^2} u_2(t) = 1 \quad \text{--- ②}$$

$$u_n''(t) + \frac{n^2\pi^2}{L^2} u_n(t) = 0, \quad n \neq 2. \quad \text{--- ③}$$

又由初始条件 ~~有~~  $u_n(0) = 0, \quad u_n'(0) = 0$

$$\therefore \text{解②式得 } u_2(t) = -\frac{L^2}{4\pi^2} \cos \frac{2\pi}{L}t + \frac{L^2}{4\pi^2}$$

$$\text{解③式得 } u_n(t) = 0, \quad n \neq 2.$$

$$\therefore u(x, t) = \frac{L^2}{4\pi^2} (1 - \cos \frac{2\pi}{L}t) \cos \frac{2\pi}{L}x$$

三. 解: 方程变为  $(dy)^2 - 5dx dy + 4(dx)^2 = 0$

$$(dy - dx)(dy - 4dx) = 0$$

特征曲线:  $\begin{cases} y - x = C_1 \\ y - 4x = C_2 \end{cases}$

特征变换:  $\begin{cases} \xi = y - x \\ \eta = y - 4x \end{cases}$

方程经特征变换化为  $\frac{\partial^2 u}{\partial \xi \partial \eta} = 0$ .

$$\therefore u = f_1(\xi) + f_2(\eta)$$

其中  $f_1, f_2$  为任意二次可微连续函数.

$$\therefore u(x, y) = f_1(y - x) + f_2(y - 4x)$$

由条件得  $\begin{cases} f_1(-x) + f_2(-4x) = 0 \dots \textcircled{1} \\ f_1'(-x) + f_2'(-4x) = 2x \dots \textcircled{2} \end{cases}$

②式积分得  $-f_1(-x) - \frac{1}{4}f_2(-4x) = x^2 + C$ .

$$\begin{cases} f_1(-x) = -\frac{4}{3}x^2 - \frac{4}{3}C \\ f_2(-4x) = \frac{4}{3}x^2 + \frac{4}{3}C \end{cases}$$

$$\begin{cases} f_1(x) = -\frac{4}{3}x^2 - \frac{4}{3}C \\ f_2(x) = \frac{1}{12}x^2 + \frac{4}{3}C \end{cases}$$

$$\begin{aligned} \therefore u(x, y) &= f_1(y - x) + f_2(y - 4x) \\ &= -\frac{4}{3}(y - x)^2 + \frac{1}{12}(y - 4x)^2 \\ &= -\frac{5}{4}y^2 + 2xy \end{aligned}$$

四.

解. 对  $x$  作傅里叶变换.

$$u(x, t) \leftrightarrow U(w, t), \quad \varphi(x) \leftrightarrow \psi(w)$$

方程化为  $\frac{\partial U}{\partial t} + w^2 U = 0$

$$\therefore U(w, t) = C \cdot e^{-w^2 t}$$

由  $U|_{t=0} = \psi(w)$  有.  $C = \psi(w)$ .

$$\therefore U(w, t) = \psi(w) \cdot e^{-w^2 t}$$

$$\therefore u(x, t) = \varphi(x) * \left( \frac{1}{\sqrt{4\pi t}} e^{-\frac{x^2}{4t}} \right)$$

$$= \int_{-\infty}^{+\infty} \frac{1}{\sqrt{4\pi t}} \varphi(x) \cdot e^{-\frac{(x-z)^2}{4t}} dz$$

五.

解: 设  $M_0(x_0, y_0, z_0)$  为半空间  $x > a$  内任意一点,

则  ~~$M_1$~~   $M_1(2a-x_0, y_0, z_0)$  是  $M_0$  关于平面  $x=a$  的对称点,

故格林函数

$$G(u, M_0) = \frac{1}{4\pi} \left( \frac{1}{r_{MM_0}} - \frac{1}{r_{MM_1}} \right)$$

$$= \frac{1}{4\pi} \left[ \frac{1}{\sqrt{(x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2}} - \frac{1}{\sqrt{(x-2a+x_0)^2 + (y-y_0)^2 + (z-z_0)^2}} \right]$$

$$\left. \frac{\partial G}{\partial n} \right|_{x=a} = - \left. \frac{\partial G}{\partial x} \right|_{x=a} = - \frac{1}{2\pi} \cdot \frac{x_0 - a}{[(a-x_0)^2 + (y-y_0)^2 + (z-z_0)^2]^{\frac{3}{2}}}$$

$$u(x_0, y_0, z_0) = - \iint_{x=a} \frac{\partial G}{\partial n} f(y, z) dy dz$$

$$= \frac{1}{2\pi} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \frac{x_0 - a}{[(a-x_0)^2 + (y-y_0)^2 + (z-z_0)^2]^{\frac{3}{2}}} f(y, z) dy dz$$

六、解  $f(x) = \sum_{i=1}^{\infty} C_i J_0(\alpha_i x)$  , 而  $f(x) = x^2$  ( $0 \leq x \leq 1$ )

$$C_i = \frac{1}{\frac{1}{2} J_1^2(\alpha_i)} \int_0^1 x \cdot x^2 J_0(\alpha_i x) dx$$

$$= \frac{2}{J_1^2(\alpha_i)} \int_0^1 x^2 \cdot x J_0(\alpha_i x) dx$$

$$\text{而 } \int_0^1 x^2 \cdot x J_0(\alpha_i x) dx$$

$$= \frac{1}{\alpha_i} x^2 \cdot x J_1(\alpha_i x) \Big|_0^1 - \frac{1}{\alpha_i} \int_0^1 2x \cdot x J_1(\alpha_i x) dx$$

$$= \frac{J_1(\alpha_i)}{\alpha_i} - \frac{2}{\alpha_i} \int_0^1 x^2 J_1(\alpha_i x) dx$$

$$= \frac{J_1(\alpha_i)}{\alpha_i} - \frac{2}{\alpha_i} \cdot \frac{1}{\alpha_i} \cdot x^2 J_2(\alpha_i x) \Big|_0^1$$

$$= \frac{J_1(\alpha_i)}{\alpha_i} - \frac{2}{\alpha_i^2} J_2(\alpha_i)$$

$$\therefore C_i = \frac{2}{\alpha_i J_1(\alpha_i)} - \frac{4 J_2(\alpha_i)}{\alpha_i^2 J_1^2(\alpha_i)}$$

$$\therefore f(x) = \sum_{i=1}^{\infty} \left[ \frac{2}{\alpha_i J_1(\alpha_i)} - \frac{4 J_2(\alpha_i)}{\alpha_i^2 J_1^2(\alpha_i)} \right] J_0(\alpha_i x)$$

$$6. \text{解} \begin{cases} \frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = 0, \quad t > 0, x > 0, x - at < 0 \\ u|_{x-at=0} = f(x) \\ u|_{x=0} = g(t) \end{cases}$$

$$\text{令} \begin{cases} \xi = x + at \\ \eta = x - at \end{cases}, \text{方程化为 } \frac{\partial^2 u}{\partial \xi \partial \eta} = 0, \text{通解为 } u(\xi, \eta) = f_1(\xi) + f_2(\eta)$$

$$\text{条件化为} \begin{cases} u|_{\eta=0} = f(\frac{\xi+\eta}{2}) = f(\frac{\xi}{2}), \text{代入通解得} \\ u|_{\eta=-\xi} = g(\frac{\xi-\eta}{2a}) = g(\frac{\xi}{a}) \end{cases}$$

$$\begin{cases} u(\xi, 0) = f_1(\xi) + f_2(0) = f(\frac{\xi}{2}) \\ u(\xi, \eta)|_{\eta=-\xi} = f_1(\xi) + f_2(-\xi) = g(\frac{\xi}{a}) \end{cases}$$

$$\text{解得} \begin{cases} f_2(-\xi) = -f(\frac{\xi}{2}) + g(\frac{\xi}{a}) + f_2(0) \Rightarrow f_2(\xi) = -f(\frac{-\xi}{2}) + g(\frac{-\xi}{a}) + f_2(0) \\ f_1(\xi) = f(\frac{\xi}{2}) - f_2(0) \end{cases}$$

$$\text{于是是解问题的解为 } u(\xi, \eta) = f(\frac{\xi}{2}) - f(\frac{-\eta}{2}) + g(\frac{-\eta}{a})$$

$$\text{原问题的解为 } u(x, t) = f(\frac{x+at}{2}) - f(\frac{at-x}{2}) + g(\frac{at-x}{a})$$